

Gradient conditions for the extreme BMO closure

FA/Banach run packet for arXiv:1409.5322

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Source question

In Remark 6.4(3) of S. Geiss and J. Ylinen, *Decoupling on the Wiener Space, Related Besov Spaces, and Applications to BSDEs*, arXiv:1409.5322, the authors ask for conditions on the BSDE gradient Z which guarantee

$$M := \int_0^\cdot c_s d\bar{W}_s \in \overline{L^\infty}^{\text{BMO}_2}. \quad (1)$$

Here c is the comparison coefficient constructed in the proof of their Theorem 6.3. Membership in (1) is the extreme case in which the Doleans–Dade exponential of M satisfies reverse Hölder inequalities for every exponent. We give two answers: a quantitative sliceability criterion and a broader exponential-energy criterion.

We use the source notation. In particular, $\text{sl}_N^{S_2, \mathcal{F}}(u)$ is the N -sliceable number of a predictable process u , and (s_N) is a non-increasing sequence such that

$$\text{sl}_N^{S_2, \mathcal{F}}(|Z|^\theta) \leq s_N, \quad s_\infty = \lim_{N \rightarrow \infty} s_N. \quad (2)$$

The generator constant is L_Z . The spaces L^∞ and H^∞ denote respectively bounded continuous martingales and continuous martingales with bounded terminal quadratic variation.

Full sufficient-condition theorem

Theorem 1. *Assume the setting and hypotheses of Theorem 6.3 of arXiv:1409.5322, with $0 \leq \theta \leq 1$, and let M be its comparison martingale.*

(i) *One always has the quantitative estimate*

$$\text{dist}_{\text{BMO}_2} \left(M, \overline{L^\infty}^{\text{BMO}_2} \right) \leq 2\sqrt{2} L_Z s_\infty. \quad (3)$$

Consequently, sliceability of $|Z|^\theta$, namely $s_\infty = 0$, implies (1).

(ii) *The conclusion holds automatically for $\theta = 0$. For $0 < \theta \leq 1$, it also holds whenever*

$$Z \in \text{bmo}(S_{2\eta}) \quad \text{for some } \eta > \theta. \quad (4)$$

More precisely,

$$\text{sl}_N^{S_2, \mathcal{F}}(|Z|^\theta) \leq \left(\frac{T}{N} \right)^{\frac{1}{2}(1-\theta/\eta)} \|Z\|_{\text{bmo}(S_{2\eta})}^\theta. \quad (5)$$

Unlike the displayed example in the source, (4) allows $\eta > 1$, and therefore includes the quadratic endpoint $\theta = 1$.

(iii) There is a sufficient condition which does not require $s_\infty = 0$. Define, for $\lambda > 0$,

$$\mathfrak{C}_\lambda(Z) := \sup_\tau \left\| \mathbb{E} \left[\exp \left(\lambda \int_\tau^T |Z_s|^{2\theta} ds \right) \middle| \mathcal{F}_\tau \right] \right\|_\infty, \quad (6)$$

where the supremum is over stopping times. If

$$\mathfrak{C}_\lambda(Z) < \infty \quad \text{for every } \lambda > 0, \quad (7)$$

then (1) holds.

The quantitative closure argument

For continuous martingales, Corollary 3 of L. Yan and N. Kazamaki, *On the distance between $\langle X \rangle$ and L^∞ in the space of continuous BMO-martingales* (2005), gives

$$\overline{H^\infty}^{\text{BMO}} \subseteq \overline{L^\infty}^{\text{BMO}}. \quad (8)$$

All martingale BMO norms are equivalent, so the closures in (8) are unchanged when the norm is BMO_2 . Lemma 5.5(4) of the source, based on Schachermayer's characterization, says

$$\text{sl}_\infty(M) = \text{dist}_{\text{BMO}_2}(M, H^\infty). \quad (9)$$

The proof of source Theorem 6.3 establishes

$$\text{sl}_{3N-2}^{S_2, \overline{\mathcal{F}}}(|c|) \leq \sqrt{2}L_Z \left[\sqrt{\frac{T}{N}} + 2s_N \right]. \quad (10)$$

The BMO norm of a stochastic integral is exactly the corresponding $\text{bmo}(S_2)$ norm of its integrand. Hence the left side of (10) is $\text{sl}_{3N-2}(M)$. Letting $N \rightarrow \infty$ and using (8)–(9) gives

$$\text{dist}_{\text{BMO}_2}(M, \overline{L^\infty}^{\text{BMO}_2}) \leq \text{dist}_{\text{BMO}_2}(M, H^\infty) = \text{sl}_\infty(M) \leq 2\sqrt{2}L_Z s_\infty,$$

which proves Theorem 1(i).

For part (ii), let $0 < \theta < \eta$, with no upper restriction on η . Conditional Holder followed by conditional Jensen gives, for a deterministic interval $I = (a, b]$,

$$\begin{aligned} & \mathbb{E} \left[\int_t^T \mathbf{1}_I(s) |Z_s|^{2\theta} ds \middle| \mathcal{F}_t \right] \\ & \leq (b-a)^{1-\theta/\eta} \left(\mathbb{E} \left[\int_t^T \mathbf{1}_I(s) |Z_s|^{2\eta} ds \middle| \mathcal{F}_t \right] \right)^{\theta/\eta}. \end{aligned}$$

Taking essential suprema and square roots, and then choosing the equidistant N -partition, proves (5). Thus $s_\infty = 0$. The $\theta = 0$ estimate $\text{sl}_N^{S_2, \overline{\mathcal{F}}}(1) \leq \sqrt{T/N}$ is immediate. This proves (ii).

A broader, direct extreme-class criterion

We prove part (iii). The canonical decoupling identities used in steps (c) and (d) of the source proof transfer (6) with the same constant to each transformed gradient Z^ρ , $\rho \in \{\varphi, \psi\}$, relative to the enlarged filtration. The source comparison coefficient satisfies

$$|c_s| \leq \sqrt{2}L_Z (1 + |Z_s^\psi|^\theta + |Z_s^\varphi|^\theta).$$

Writing $Q_\tau = \int_\tau^T |c_s|^2 ds$, we therefore have

$$Q_\tau \leq 6L_Z^2 \left[T + \int_\tau^T |Z_s^\psi|^{2\theta} ds + \int_\tau^T |Z_s^\varphi|^{2\theta} ds \right]. \quad (11)$$

Conditional Cauchy–Schwarz and (7) imply, for every $\lambda > 0$,

$$\sup_\tau \left\| \mathbb{E}[e^{\lambda Q_\tau} \mid \mathcal{F}_\tau] \right\|_\infty \leq e^{6\lambda L_Z^2 T} \mathfrak{C}_{12\lambda L_Z^2}(Z) < \infty. \quad (12)$$

In particular $M \in \text{BMO}$.

Let $R = \mathcal{E}(M)$ and fix $q > 1$. Since $2qM \in \text{BMO}$, its stochastic exponential is a uniformly integrable martingale. For every stopping time τ , Cauchy–Schwarz yields

$$\begin{aligned} \mathbb{E} \left[\left(\frac{R_T}{R_\tau} \right)^q \middle| \mathcal{F}_\tau \right] &= \mathbb{E} \left[e^{q(M_T - M_\tau) - \frac{q}{2} Q_\tau} \middle| \mathcal{F}_\tau \right] \\ &\leq \mathbb{E} \left[\frac{\mathcal{E}(2qM)_T}{\mathcal{E}(2qM)_\tau} \middle| \mathcal{F}_\tau \right]^{1/2} \mathbb{E} \left[e^{(2q^2 - q)Q_\tau} \middle| \mathcal{F}_\tau \right]^{1/2} \\ &\leq \left[e^{6(2q^2 - q)L_Z^2 T} \mathfrak{C}_{12(2q^2 - q)L_Z^2}(Z) \right]^{1/2} < \infty. \end{aligned}$$

Thus $R \in \text{RH}_q$ for every $q > 1$. Kazamaki’s extreme-class characterization, quoted immediately after Proposition 5.24 in the source, is precisely that this is equivalent to $M \in \overline{L^\infty}^{\text{BMO}_2}$. This proves (iii).

Strength, boundary, and novelty

Condition (7) genuinely reaches beyond the sliceable class. Schachermayer’s 1996 example has infinite conditional quadratic-variation critical exponent, hence the every-rate property, but is not in the BMO closure of H^∞ . The new direct argument only seeks the larger L^∞ -closure requested by the source.

An exact necessary condition expressed solely through $|Z|$ cannot hold uniformly over all generators: if the generator is independent of its z -variable, then $c = 0$ and (1) holds for every admissible Z . The theorem therefore answers the source’s request at its natural level—broad, checkable sufficient conditions—and adds a quantitative distance estimate.

A bounded search on 13 August 2026 used the exact source title and arXiv id, the quoted future-work phrase, both authors, and combinations of gradient, sliceability, BMO closure, and reverse Holder terms. It found the source, the same authors’ related weighted-BMO paper arXiv:1501.01183, and the classical ingredients, but no posted follow-up stating either result above.

References

- S. Geiss and J. Ylinen, *Decoupling on the Wiener Space, Related Besov Spaces, and Applications to BSDEs*, Mem. Amer. Math. Soc. 272 (2021), no. 1335; arXiv:1409.5322.
- L. Yan and N. Kazamaki, *On the distance between $\langle X \rangle$ and L^∞ in the space of continuous BMO-martingales*, Studia Math. 168 (2005), 129–134.
- W. Schachermayer, *A characterisation of the closure of H^∞ in BMO*, Seminaire de Probabilites XXX, Lecture Notes in Math. 1626 (1996), 344–356.
- S. Geiss and J. Ylinen, *Weighted Bounded Mean Oscillation applied to Backward Stochastic Differential Equations*, arXiv:1501.01183 (2015).

Human-review recommendation. Review as a short full sufficient-condition result. The key checks are the closure inclusion (8), the limit in (10), the unrestricted $\eta > \theta$ Holder step, and the exponent $2q^2 - q$ in the direct reverse-Holder calculation.